

210 HW #8

1. Flow chart & DOF ^{1st Ind DOF (I don't eat falafel) (for lunch today)}

$$DOF = 23 + 2 - (2 + 18 + 2 + 4) = 0$$

b.) $n_{CO_2} = \frac{45000 \times 4.3}{28} = 803.57$, $n_{O_2} = \frac{45000 \times 18.2}{44} = 204.55$, $n_{N_2} = \frac{45000 \times 75.7}{28} = 482.14$ $\frac{K_{H_2O}}{K_{H_2O}}$

$\rightarrow 0.8434 \times 803.57 = 678.42 = 6 \text{ } \frac{K_1}{K_2} + 4 \text{ } \frac{K_2}{K_2} = 6 \left(\frac{21000 \times 0.17}{46} \right) + 4 \text{ } \frac{K_2}{K_2} \rightarrow \frac{K_1}{K_2} = 86.96, \frac{K_2}{K_2} = 39.17$

$n_{CO_2} = 803.57 - 678.42 = 125.15$, $H_2O = 4 \text{ } \frac{K_1}{K_2} + \frac{K_2}{K_2} = 39.17$

$n_{O_2} = 204.55 + 426.18 = 630.73$ $\frac{K_{H_2O}}{K_{H_2O}}$, $n_{H_2O} = \frac{32000 \times 4000}{18} = 155.56$ $\frac{K_{H_2O}}{K_{H_2O}}$

$\rightarrow$ This is a lot to write out, I'm just gonna do this through excel & make a table

Stream	M	D	N	W	A	E
1	803.57	204.55	482.14	—	—	—
2	—	—	—	31858.4	—	—
3	803.57	204.55	482.14	6400.5	21.54	—
4	125.15	630.73	482.14	—	—	—
5	125.15	630.73	482.14	68141.3	39.17	86.96
6	—	—	—	155.56	—	86.96
7	—	—	—	66545.7	39.17	—
8	—	—	—	29963.6	17.63	—
9	—	—	—	36622.1	21.54	—

excel file?
—15

(c.) To improve efficiency, reducing the purge rate allows more water recycled, assuming the extra acetic acid isn't an issue

conversions?
—10

25150

5.34 ▶ 30% EtOH & 70 mol% H₂O, 1 atm

$F_{\text{tot}} = \text{Vapor} + \text{liquid}$, $F_{\text{tot}} \text{ mol: } F_{\text{Et}} = V_{\text{Et}} + L_{\text{Et}} \rightarrow \text{For } 50/50 \rightarrow z_{\text{Et}} = \frac{1}{2} y_{\text{Et}} + \frac{1}{2} x_{\text{Et}}$

$\rightarrow 2z_{\text{Et}} = y_{\text{Et}} + x_{\text{Et}}$ with feed, $x_{\text{Et}} + y_{\text{Et}} = 0.6 \rightarrow \text{from b.11, } 85.3^\circ \rightarrow x_{\text{Et}} + y_{\text{Et}} = 0.5942$
 $, 84.1^\circ = 0.6790 \rightarrow \frac{0.6 - 0.5942}{0.679 - 0.5942} = 0.0718 \rightarrow T_{\text{b.p.}}(85 + 0.0718)(84.1 - 85.3) = \boxed{85.2^\circ \text{C}}$

Liquid Phase: $(x_{\text{Et}} = (0.1238 + 0.0718)(0.1161 - 0.1238) = 0.1268 \rightarrow 1 - 0.1268 = 0.8732$

$\rightarrow \text{Liquid} = 12.68\% \text{ EtOH, } 87.32\% \text{ H}_2\text{O}$

Vapor Phase: $V_{\text{Et}} = (0.4704 + 0.0718)(0.5089 - 0.4704) = 0.4732, 1 - 0.4732 = 0.5268$

$\rightarrow \text{Vapor} = 47.32\% \text{ EtOH, } 52.68\% \text{ H}_2\text{O}$

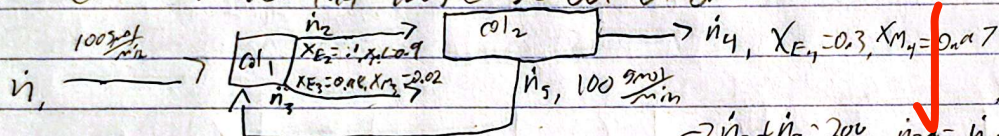
Fractional Recoveries:

Vapor Recovery $= R_{\text{Et,v}} = \frac{V_{\text{Et}}}{F_{\text{Et}}} = \frac{\frac{1}{2} 0.4732}{0.3} = 0.7887 = 78.87\% \text{ EtOH Recovery in Vapor}$

Liquid Recovery $H_2O = R_{\text{H}_2\text{O,l}} = \frac{L_{\text{H}_2\text{O}}}{F_{\text{H}_2\text{O}}} = \frac{\frac{1}{2} 0.5268}{0.7} = 0.6257 = 62.57\% \text{ H}_2\text{O Recovery in liquid}$

Separation Factor $= \alpha = \frac{\frac{y_{\text{Et}}}{x_{\text{Et}}}}{\frac{y_{\text{H}_2\text{O}}}{x_{\text{H}_2\text{O}}}} = \frac{0.4732/0.5268}{0.1268/0.8732} = 6.18$

5.32 ▶ You do not notice that these are out of order.



$\rightarrow \dot{n}_2 + \dot{n}_3 = 200, \dot{n}_2 = \dot{n}_4 + 100$

$\rightarrow 0.3(100) = 0.98(\dot{n}_2) + 0.3(\dot{n}_4), \dot{n}_2 = 30.67 = 0.0306 \dot{n}_4$

$0.7(100) = 0.02 - (0.02 \dot{n}_4) + 0.97 \dot{n}_4 \rightarrow \dot{n}_4 = 71.579, \dot{n}_2 = 28.41$

$200 = \dot{n}_2 + 28.41 \rightarrow \dot{n}_2 = 171.59$

$\dot{n}_2 = \dot{n}_4 + \dot{n}_5 \rightarrow 171.59 = 71.579 + 100, \dot{n}_{\text{Et}} = 1.59$

$X_{\text{Et}}(100) + 30 = 0.1(171.59) + 0.98(28.41) = 0.15 = X_{\text{Et}}, 1 - 0.15 = 0.85 = X_{\text{H}_2\text{O}}$

$X_{\text{Et}}(100) + 0.7(100) = 0.1(171.59) + 0.97(28.41) = 0.15 = X_{\text{Et}}, 1 - 0.15 = 0.85 = X_{\text{H}_2\text{O}}$

Fractional Recovery!

$E = \frac{0.3(71.59) + 0.98(28.41)}{0.3(100)} = 0.9996 = 99.96\%$

$M = \frac{0.97(171.59) + 0.02(28.41)}{0.7(100)} = 0.9965 = 99.65\%$